

BS Artificial Intelligence (AI) Program

This document outlines the BS Artificial Intelligence (AI) Program structure, Professional Educational Objectives (PEOs), Program Learning Outcomes (PLOs), and admission criteria. It is formatted for clarity and easy reference.

Introduction

The BS Artificial Intelligence program is designed to empower students with the skills and knowledge needed to excel in the rapidly evolving fields of machine learning, robotics, and language processing. Students will develop the expertise to create innovative, AI-driven solutions for real-world challenges. The curriculum provides a comprehensive understanding of AI theories, methods, and technologies, focusing on technical, analytical, critical-thinking, and ethical practices. Graduates will be prepared to develop and implement AI systems across various interdisciplinary domains as responsible AI practitioners.

Professional Educational Objectives (PEOs)

The AI Graduates will be able to:

Apply Principles: Demonstrate the ability to apply the principles, methods, tools, and technologies of Artificial Intelligence in analyzing, designing, developing, and formulating solutions for complex computing problems, and make decisions based on data and information.

Ethics and Teamwork: Build ethical and legal responsibilities while working effectively as individuals and in interdisciplinary teams, along with the ability to communicate technical information effectively to both technical and non-technical audiences.

Lifelong Learning: Utilize the lifelong learning ability and professional capabilities to comprehend and comply with the continuously evolving Artificial Intelligence domain and apply the latest trends and knowledge to evolve the professional society.

Program Learning Outcomes (PLOs)

Drawing from the Seoul Accord's internationally recognized standards, the program focuses on:

PLO #	Learning Outcome Title
-------	------------------------

PLO #	Learning Outcome Title
PLO1	Academic Education
PLO2	Knowledge for Solving Computing Problems
PLO3	Problem Analysis
PLO4	Design/Development of Solutions
PLO5	Modern Tool Usage
PLO6	Individual and Team Work
PLO7	Communication
PLO8	Computing Professionalism and Society
PLO9	Ethics
PLO10	Life-long Learning

Graduates can pursue careers as:

AI/ML Engineer

Data Scientist

Intelligent Automation Engineer

AI Software Developer

Natural Language Processing Engineer

AI Product Manager

AI Consultant

AI Research Scientist

Semester 1 (Total Cr. Hr: 19)

Course Code	Course Title	Prerequisite	Cr. Hr.
-------------	--------------	--------------	---------

Course Code	Course Title	Prerequisite	Cr. Hr.
CC111	Programming Fundamentals	-	3
CC111L	Programming Fundamentals Lab	-	1
MA107	Calculus and Analytical Geometry	-	3
CC120	Application of Information & Communication Technologies	-	2
CC120L	Application of Information & Communication Technologies lab	-	1
EN110	English I	-	3
NS125	Applied Physics	-	2
NS125L	Applied Physics Lab	-	1
POL121	Pakistan: Ideology, Constitution, and Society	-	3

Semester 2 (Total Cr. Hr: 19)

Course Code	Course Title	Prerequisite	Cr. Hr.
ISL112	Islamic Thought and Perspectives	-	3
EN123	English II	EN110	3
MA108	Multivariable Calculus (AI Supporting 1)	MA107	3
CC112	Object Oriented Programming	CC111, CC111L	3
CC112L	Object Oriented Programming	CC111,	1

Course Code	Course Title	Prerequisite	Cr. Hr.
	Lab	CC111L	
CC121	Digital Logic Design	-	2
CC121L	Digital Logic Design Lab	-	1
CC141	Discrete Structures	-	3

Semester 3 (Total Cr. Hr: 18)

Course Code	Course Title	Prerequisite	Cr. Hr.
CC213	Data Structures	CC112, CC112L	3
CC213L	Data Structures Lab	CC112, CC112L	1
CC281	Software Engineering	-	3
MA150	Probability & Statistics (AI Supporting 3)	-	3
MA210	Linear Algebra (AI Supporting 2)	MA107	3
CC222	Computer Organization & Assembly Language	CC121, CC121L	2
CC222L	Computer Organization & Assembly Language Lab	CC121, CC121L	1
SD210	Civics and Community Engagement	-	2

Semester 4 (Total Cr. Hr: 19)

Course Code	Course Title	Prerequisite	Cr. Hr.
CC251	Computer Networks	-	2
CC251L	Computer Networks Lab	-	1
CC230	Database Systems	-	3
CC230L	Database Systems Lab	-	1
AI271	Programming for AI	-	2
AI271L	Programming for AI Lab	-	1
TE 1	Technical Elective 1 (e.g., Open-Source Software Development)	-	3
UE 1	University Elective 1 (Arts and Humanities)	-	3
HU201	Professional Practices	-	3

Semester 5 (Total Cr. Hr: 19)

Course Code	Course Title	Prerequisite	Cr. Hr.
CC323	Operating Systems	-	3
CC323L	Operating Systems Lab	-	1
CC371	Artificial Intelligence	CC213, CC213L	3
CC342	Analysis of Algorithms	CC213, CC213L	3
MG224	Innovation and Entrepreneurship	-	3

Course Code	Course Title	Prerequisite	Cr. Hr.
TE 2	Technical Elective 2	-	3
AI372	Machine Learning	AI271, AI271L	2
AI372L	Machine Learning Lab	AI271, AI271L	1
SD100	English Immersion	-	0

Semester 6 (Total Cr. Hr: 18)

Course Code	Course Title	Prerequisite	Cr. Hr.
TE 3	Technical Elective 3	-	3
CC361	Information Security	-	3
SD102	21st Century	-	0
AI374	Knowledge Representation & Reasoning	AI271, AI271L	3
AI473	Artificial Neural Networks & Deep Learning	AI372, AI372L	2
AI473L	Artificial Neural Networks & Deep Learning Lab	AI372, AI372L	1
TE 4	Technical Elective 4	-	3
EN220	Technical & Business Writing (AI Supporting 4)	EN125	3

Semester 7 (Total Cr. Hr: 14)

Course Code	Course Title	Prerequisite	Cr. Hr.
-------------	--------------	--------------	---------

Course Code	Course Title	Prerequisite	Cr. Hr.
AI375	Computer Vision	AI473, AI473L	2
AI375L	Computer Vision Lab	AI473, AI473L	1
TE 5	Technical Elective 5	-	3
TE 6	Technical Elective 6	-	3
CC425	Parallel and Distributed Computing	CC323, CC323L	3
CC491	Final Year Project I / COOP 1*	-	2

*COOP means FYP undertaken in the industry.

Semester 8 (Total Cr. Hr: 10)

Course Code	Course Title	Prerequisite	Cr. Hr.
UE 2	University Elective 2 (Social Sciences)	-	3
TE 7	Technical Elective 7	-	3
CC492	Final Year Project II / COOP 2*	CC491	4

Technical & University Electives

Course Code	Course Title	Cr. Hr.
AI391	Natural Language Processing	3
AI392	Speech Processing	3
DS393	Data Mining	\$2+1\$

Course Code	Course Title	Cr. Hr.
AI393	Reinforcement Learning	3
AI394	Large Language Models	3
AI376	Fuzzy Systems	3
AI377	Introduction to Robotics	\$2+1\$
AI378	Agent Based Modeling	3
CS393	Mobile Application Development	\$2+1\$
IT291	Web Technologies	\$2+1\$
AI379	Topics in AI	3
DS391	Big Data Analytics	3
SE492	Open-Source Software Development	\$2+1\$
IT352	Internet of Things	3
IT461	Blockchain Technology and Application	3

University Elective 1 - Arts and Humanities (3 Credit Hours) - Minimum One Course

Course Code	Course Title	Cr. Hr.
SD221	Life and Learning	3
SD222	Foreign Language	3
SD223	Logical Reasoning	3

University Elective 2 - Social Sciences (3 Credit Hours) - Minimum One Course

Course Code	Course Title	Cr. Hr.
-------------	--------------	---------

Course Code	Course Title	Cr. Hr.
HWB101	Health and Wellbeing	3
SOC102	Understanding Society and Sociocultural Dynamics	3
PSY110	Psychology of Mind and Behavior	3
ECO202	Economics	3
MK210	Principles of Marketing	3
MG120	Principles of Management	3

General Requirements

Minimum 50% marks in Intermediate/12 years of education/A- Level (HSSC) or Equivalent, with Mathematics are required for admission in all BS Computing Programs.

Equivalency certificate by IBCC will be required for education from other countries or systems.

Pre-Medical Candidates

Minimum 50% marks in Intermediate Pre-medical (subject combination of Physics, Chemistry, and Biology) or equivalent 12 years of education are also eligible to apply.

Such candidates, if admitted, are required to complete two additional Mathematics courses (worth 6 credit hours) during the first year of their studies.

This document outlines the BS Data Science (DS) Program structure, Professional Educational Objectives (PEOs), Program Learning Outcomes (PLOs), career prospects, and admission criteria. The fee structure for various programs has also been included.

BS DATA SCIENCE

Introduction

The BS Data Science program is designed to shape forward-thinking professionals capable of driving innovation through data. It provides a strong foundation in statistical methods, programming, and big data technologies. The curriculum integrates rigorous academic content with applied learning, spanning mathematics, machine learning, data engineering, and visualization to build a versatile skill set. Graduates are prepared to efficiently manage large datasets, extract actionable insights through mining, cleaning, and visualization, and solve complex, real-world problems.

Professional Educational Objectives (PEOs)

The DS Graduates will be able to:

Apply and Design Skills: Demonstrate the ability to apply and design DS acquired skills in data-driven analysis, data modeling, and providing holistic solutions to meet organizational needs across a variety of industries, both domestically and globally.

Professional Conduct: Develop lofty professional standards, ethics, communication skills, and societal concerns, enabling a graduate to work effectively as a member or team leader for their professional pursuits and communities.

Lifelong Growth: Improve professional growth through lifelong learning activities such as research, advancement updates, learning, and continuing higher education.

Program Learning Outcomes (PLOs)

Drawing from the Seoul Accord's internationally recognized standards, the program focuses on:

PLO #	Learning Outcome Title
PLO1	Academic Education
PLO2	Knowledge for Solving Computing Problems

PLO #	Learning Outcome Title
PLO3	Problem Analysis
PLO4	Design/Development of Solutions
PLO5	Modern Tool Usage
PLO6	Individual and Team Work
PLO7	Communication
PLO8	Computing Professionalism and Society
PLO9	Ethics
PLO10	Life-long Learning

Graduates can pursue careers as:

Data Scientist

Data Analyst

Machine Learning Engineer

Business Intelligence Analyst

Data Engineer

AI Engineer

Big Data Analyst

Quantitative Analyst

Semester 1 (Total Cr. Hr: 19)

Course Code	Course Title	Prerequisite	Cr. Hr.
CC111	Programming Fundamentals	-	3
CC111L	Programming Fundamentals Lab	-	1

Course Code	Course Title	Prerequisite	Cr. Hr.
MA107	Calculus and Analytical Geometry	-	3
CC120	Application of Information & Communication Technologies	-	2
CC120L	Application of Information & Communication Technologies Lab	-	1
EN110	English I	-	3
NS125	Applied Physics	-	2
NS125L	Applied Physics Lab	-	1
POL121	Pakistan: Ideology, Constitution, and Society	-	3

Semester 2 (Total Cr. Hr: 19)

Course Code	Course Title	Prerequisite	Cr. Hr.
ISL112	Islamic Thought and Perspectives	-	3
EN123	English II	EN110	3
MA108	Multivariable Calculus (DS Supporting 1)	MA107	3
CC112	Object Oriented Programming	CC111, CC111L	3
CC112L	Object Oriented Programming Lab	CC111, CC111L	1
CC121	Digital Logic Design	-	2

Course Code	Course Title	Prerequisite	Cr. Hr.
CC121L	Digital Logic Design Lab	-	1
CC141	Discrete Structures	-	3

Semester 3 (Total Cr. Hr: 18)

Course Code	Course Title	Prerequisite	Cr. Hr.
CC213	Data Structures	CC112, CC112L	3
CC213L	Data Structures Lab	CC112, CC112L	1
MA210	Linear Algebra (DS Supporting 2)	MA107	3
SS103	Civics and Community Engagement	-	2
CC222	Computer Organization and Assembly Language	CC121, CC121L	2
CC222L	Computer Organization and Assembly Language Lab	CC121, CC121L	1
CC281	Software Engineering	-	3
MA150	Probability and Statistics (DS Supporting 3)	-	3

Semester 4 (Total Cr. Hr: 19)

Course Code	Course Title	Prerequisite	Cr. Hr.
CC251	Computer Networks	-	2

Course Code	Course Title	Prerequisite	Cr. Hr.
CC251L	Computer Networks Lab	-	1
CC230	Database Systems	-	3
CC230L	Database Systems Lab	-	1
TE 1	Technical Elective 1	-	3
HU201	Professional Practices	-	3
UE 1	University Elective 1 (Arts and Humanities)	-	3
DS291	Introduction to Data Science	MA150	2
DS291L	Introduction to Data Science Lab	MA150	1

Semester 5 (Total Cr. Hr: 19)

Course Code	Course Title	Prerequisite	Cr. Hr.
CC323	Operating Systems	-	3
CC323L	Operating Systems Lab	-	1
TE 2	Technical Elective 2	-	3
SD100	English Immersion	-	0
MG365	Innovation and Entrepreneurship	-	3
DS393	Data Mining	DS291, DS291L	2
DS393L	Data Mining Lab	DS291, DS291L	1

Course Code	Course Title	Prerequisite	Cr. Hr.
CC342	Analysis of Algorithms	CC213, CC213L	3
CC371	Artificial Intelligence	CC213, CC213L	3

Semester 6 (Total Cr. Hr: 18)

Course Code	Course Title	Prerequisite	Cr. Hr.
TE 3	Technical Elective 3	-	3
CC361	Information Security	-	3
DS392	Advanced Statistics	MA150	3
SD101	21st Century	-	0
TE 4	Technical Elective 4	-	3
EN220	Technical & Business Writing	EN107	3
DS395	Data Warehousing & Business Intelligence	DS393, DS393L	3

Semester 7 (Total Cr. Hr: 14)

Course Code	Course Title	Prerequisite	Cr. Hr.
TE 5	Technical Elective 5	-	3
CC425	Parallel & Distributed Computing	CC323, CC323L	3
TE 6	Technical Elective 6	-	3

Course Code	Course Title	Prerequisite	Cr. Hr.
DS494	Data Visualization	DS395	2
DS494L	Data Visualization Lab	DS395	1
CC491	Final Year Project I / COOP 1*	-	2

*COOP means FYP undertaken in the industry.

Semester 8 (Total Cr. Hr: 10)

Course Code	Course Title	Prerequisite	Cr. Hr.
UE 2	University Elective 2 (Social Sciences)	-	3
TE 7	Technical Elective 7	-	3
CC492	Final Year Project II / COOP 2*	CC491	4

Technical & University Electives

Course Code	Course Title	Cr. Hr.
AI372	Machine Learning	\$2+1\$
IT461	Blockchain Technology and Application	3
DS391	Big Data Analytics	3
AI473	Artificial Neural Networks & Deep Learning	\$2+1\$
CS324	Human Computer Interaction	3
AI394	Large Language Models	3
DS492	Topics in Data Science	3

Course Code	Course Title	Cr. Hr.
AI377	Introduction to Robotics	\$2+1\$
AI374	Knowledge Representation & Reasoning	3
AI391	Natural Language Processing	3
AI378	Agent-Based Modeling	3
CS393	Mobile Application Development	\$2+1\$
IT291	Web Technologies	\$2+1\$
IT352	Internet of Things	3
CY361	Cyber Security	\$2+1\$
SE492	Open-Source Software Development	\$2+1\$

University Elective 1 - Arts and Humanities (3 Credit Hours) - Minimum One Course

Course Code	Course Title	Cr. Hr.
SD221	Life and Learning	3
SD222	Foreign Language	3
SD223	Logical Reasoning	3

University Elective 2 - Social Sciences (3 Credit Hours) - Minimum One Course

Course Code	Course Title	Cr. Hr.
HWB101	Health and Wellbeing	3
SOC102	Understanding Society and Sociocultural Dynamics	3

Course Code	Course Title	Cr. Hr.
PSY110	Psychology of Mind and Behavior	3
ECO202	Economics	3
MK210	Principles of Marketing	3
MG120	Principles of Management	3

General Requirements

Minimum 50% marks in Intermediate/12 years of education/A- Level (HSSC) or Equivalent, with Mathematics are required for admission in all BS Computing Programs.

Equivalency certificate by IBCC will be required for education from other countries or systems.

Pre-Medical Candidates

Minimum 50% marks in Intermediate Pre-medical (subject combination of Physics, Chemistry, and Biology) or equivalent 12 years of education are also eligible to apply.

Such candidates, if admitted, are required to complete two additional Mathematics courses (worth 6 credit hours) during the first year of their studies.

FEE STRUCTURE

Title	Degree	Duration	Fee	Program Type	Degree Level
BS Artificial Intelligence	BSAI	4 years	1,890,000	Morning	Undergraduate Programs

Title	Degree	Duration	Fee	Program Type	Degree Level
e					
BS Blockchain Technology	BSBC T	4 Years	1,690,000	Morning	Undergraduate Programs
BS Computer Science	BSCS	4 years	2,090,000	Morning	Undergraduate Programs
BS Computer Science - Post ADP	BSCS Post ADP	2 years	1,191,303	Morning	Post Associate Degree Programs
BS Cyber Security	BS CySec	4 Years	1,690,000	Morning	Undergraduate Programs
BS Data Science	BSDS	4 Years	1,690,000	Morning	Undergraduate Programs
BS Gaming and Immersive Media	BSGIM	4 Years	1,690,000	Morning	Undergraduate Programs
BS Information Technology	BSIT	4 years	1,690,000	Morning	Undergraduate Programs
BS Software Engineeri	BSSE	4 years	1,990,000	Morning	Undergraduate Programs

Title	Degree	Duration	Fee	Program Type	Degree Level
ng					
MS Artificial Intelligence	MSAI	2 Years	690,000	Weekend	Graduate Programs
MS Computer Science	MSCS	2 Years	690,000	Weekend	Graduate Programs
MS Data Science	MSDS	2 Years	690,000	Weekend	Graduate Programs
MS Information Security	MSIS	2 years	690,000	Weekend	Graduate Programs
MS Information Technology	MSIT	2 Years	690,000	Weekend	Graduate Programs
MS Software Engineering	MSSE	2 Years	690,000	Weekend	Graduate Programs
PhD Computer Science	PhD CS	3 Years	990,000	Weekend	PhD Programs